

Homework 3

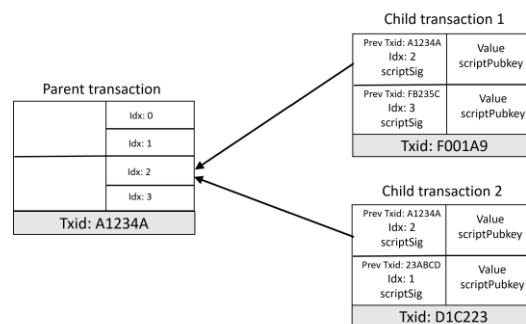
Deadline: 11:55 pm April 12, 2026

General Instructions: You may partner with up to three other students (**a group of max 4**) to submit a single write-up in a single group. We encourage discussion with other students, even if they are not in your group. Each student must build a good understanding of all the questions even if they discuss and collaborate with others. Merely dividing the problems among group members and putting those together before submission will severely impact your learning, and we **strongly advise against it**. Furthermore, while it is okay to use online resources, including **Generative AI** such as **ChatGPT**, for learning purposes, **blatant copying** or **mindlessly querying** the Generative AI is **forbidden**. If there is any confusion or questions, post those on Whatsapp group or see your TA or Instructor.

Submission Instructions: Typed or hand-written solutions (only in PDF format) can be submitted on LMS before the deadline. Email submissions WILL NOT be accepted. There will be NO late days or extensions. *Only one submission per group is needed; the first page must clearly mark the names and students IDs of all group members.*

Question 1 [7 marks]

Consider the following image:



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Are both of the transactions valid? Why and why not? If any of the two transactions is invalid, explain how the UTXO model inherently prevents this problem.

Question 2 [10 marks]

Explain how multi-signature (multisig) transactions are created in the Bitcoin network. How did Segwit improve the handling of multi-sig transactions? Discuss the differences in transaction size and verification process. Also explain how segwit improved on the security of multisig transactions.

Question 3 Segwit Multisig Account [24 pts]:

Eman, Namwar, and Zaeem have locked 3 BTC in 'eman-namwar-zaeem' joint account by creating a segwit 'pay to witness script hash' (p2wsh) transaction. Assume that the txid of the locking transaction is TXABC123 and it has two outputs. The funds are locked at the first output (index 0), while a 32 B hash of an important document called 'Project Sigma' is stored at the second output (index 1).

- a)** Let <sh> denote the script hash and <psigh> denote the hash of Project Sigma. Write down the complete scripts for the two outputs of this transaction. (4 pts)
- b)** What will be the increase in UTXO database size once the txid TXABC123 is confirmed and added to the blockchain? (3 pts)
- c)** What is the advantage of storing the hash of Project Sigma on the blockchain? (2 pts)
- d)** Suppose Eman and Namwar have to pay 0.8 BTC to Sana. The spending rules of txid TXABC123 require 2 of 3 signatures. Eman and Namwar create a new segwit spending txn with txid TXDEF456, which pays 0.8 BTC to Sana and locks the remaining funds in the 'eman-namwar-zaeem' joint account with the same spending rules as txid TXABC123. Write the complete scripts of the input including the witness field and the scripts of all the outputs of txid TXDEF456. Assume <sigE> and <sigN> represent the signatures of Eman and Namwar and <pkE>, <pkN>, and <pkZ> are the public keys of Eman, Namwar and Zaeem. The pubkey hash of Sana is <pkhS>. Assume the transaction fee is 0.1 BTC. (5 pts)
- e)** Suppose a segwit node wants to validate txid TXDEF456. Assuming that the transaction is valid, explain all the steps involved in the validation procedure for the segwit node. (3 pts)
- f)** Suppose a non-segwit node (old node) wants to validate txid TXDEF456. Assuming that the transaction is valid, explain all the steps involved in the validation procedure for the old non-segwit node. (3 pts)
- g)** Suppose Sana knows about the transaction malleability vulnerability of Bitcoin transactions. She also knows a corrupt miner who can malleate transactions. Will Sana and her miner friend succeed in malleating txid TXDEF456? Explain your answer. (2 pts)
- h)** Suppose Zain knows the spending rules of 'eman-namwar-zaeem' joint account and also knows all the public keys. With this information, Zain can create a serialized script whose hash equals the script hash <sh> at output 0 of txid TXABC123. Assuming that Zain is malicious, can he create a transaction to unlock the funds in the joint account? (2 pts)

Question 4 Multiple Transactions [12 pts] Suppose the following transactions are broadcasted **simultaneously** on the blockchain network and all become available to a miner.

Transaction hash: ABC3

Inputs	Outputs
Some valid input worth 10 BTC	Alice → 3.95 BTC
	Bob → 1.95 BTC
	Charlie → 3.90 BTC

Transaction hash: BD9E

Inputs	Outputs
Prev hash: ABC3 Idx: 0 Signed by Alice	Dianne → 3.9 BTC

Transaction hash: C4B7

Inputs	Outputs
Prev hash: ABC3 Idx: 1 Signed by Bob	Fring → 1.90 BTC
Prev hash: ABC3 Idx: 2 Signed by Charlie	Bob → 1.0 BTC
	Charlie → 2.85 BTC

Transaction hash: D743

Inputs	Outputs
Prev hash: ABC3 Idx: 0 Signed by Alice	George → 4.0 BTC
Prev hash: C4B7 Idx: 2 Signed by Charlie	Hank → 2.60 BTC

The miner has the flexibility to pick the transactions in any order. What transaction order would maximize their mining fee? What would be that fee?

a) Identify all the invalid transactions (you should mention the transaction hashes, which are given). Also state a reason, why do you think a transaction is invalid (6 pts).

b) Suppose the valid transactions make it to the mempool and a miner picks them up for inclusion in a block. How much fee would he receive from these transactions. Show your working (6 pts).

Question 5 Advanced SegWit Concepts [15 pts]

Segregated Witness (SegWit) was a major upgrade to Bitcoin introduced via a soft fork in 2017. It changed how transaction data is organized and validated, enabling scalability improvements and new features.

a) Explain how SegWit changes the structure of a Bitcoin transaction compared to a legacy transaction. Specifically, describe what is stored in the *witness* field and how it relates to the scriptSig and scriptPubKey (5 pts).

b) Before SegWit, Bitcoin transactions were vulnerable to a form of *transaction malleability*. Explain what transaction malleability is, how it could be exploited, and how SegWit solves this problem. (5 pts)

c) SegWit was deployed as a **soft fork** rather than a hard fork. Explain how this was achieved in terms of transaction compatibility. In your answer, describe how older nodes (that have not upgraded) interpret SegWit transactions and why consensus is still maintained. (5 pts)

Question 6 Merkle Tree and Hashing [11 pts]

A Bitcoin block contains 2048 transactions. A lightweight SPV (Simplified Payment Verification) client wants to verify that a specific transaction is included in this block.

a) Calculate the exact number of hashes the SPV client needs to receive as proof to verify the transaction's inclusion in the block. Show your calculation. [3 pts]

b) If the block size increases to contain 16,384 transactions, how many additional hashes would be needed for the Merkle proof compared to part (a)? Explain why Merkle trees provide logarithmic scalability. [4 pts]

c) Suppose an attacker tries to create a fake Merkle proof by modifying transaction Tx₅ in a block containing 8 transactions. Draw the Merkle tree and mark with an 'X' all the nodes that would have different hash values due to this modification. Explain why this makes fraud detection efficient. [4 pts]

Question 7 UTXO Model [12 pts]:

Consider the following scenario involving Alice, Bob, and Charlie:

- UTXO₁: Alice owns 5 BTC at (Txid: XYZ789, Index: 0)
- UTXO₂: Alice owns 3 BTC at (Txid: ABC456, Index: 1)
- UTXO₃: Bob owns 2 BTC at (Txid: DEF123, Index: 0)

a) Alice wants to pay Bob 6 BTC and Charlie 1.5 BTC in a single transaction. Design a complete valid transaction showing all inputs, outputs, and transaction fee. Explain why Alice needs to use both UTXOs and what happens to remaining funds. [3 pts]

b) After the transaction in part (a) is confirmed, list all UTXOs added to and removed from the UTXO database. [2 pts]

c) Bob now wants to create two simultaneous transactions: T1 sending 3 BTC to Charlie, and T2 sending 3 BTC to Alice. Both transactions reference Bob's UTXO from Alice's payment in part (a), which contains 6 BTC at (NEW001, 0). Are both transactions valid or invalid (explain), what happens when they are broadcast to the network, and which transaction gets confirmed (what are the determining factors behind a transaction getting confirmed). [5 pts]

d) Explain why the UTXO model prevents double-spending better than an account-based model. [2pts]

Question 8 Merkle Trees and UTXO [13 pts]:

Bitcoin's design uses Merkle trees in block headers to commit to transactions, while nodes maintain a separate UTXO database.

a) Explain why Bitcoin uses two separate data structures (Merkle tree of transactions in blocks and UTXO database) rather than just storing all unspent outputs in a Merkle tree in the block header. Discuss at least two practical reasons related to efficiency and functionality. [5 pts]

b) An SPV client receives a Merkle proof showing that transaction Tx_K is in Block 800,000. However, this SPV client later wants to verify that Tx_K hasn't been spent yet. Explain why the Merkle proof alone is insufficient for this verification, and what additional information the client would need. What security assumption must the SPV client make? [8 pts]